

MOYALE, Kenya

WMO: 636190

Lat: 3.533N

Lon: 39.050E

Elev: 1097

StdP: 88.82

Time Zone: 3.00 (E03)

Period: 90-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	16.9	2.5	5.2	27.7	4.5	5.9	26.9	11.0	20.5	9.8	20.7	3.3	180	0.285

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.1	32.7	17.7	31.9	17.7	31.0	17.8	22.0	25.9	21.6	25.5	21.2	25.1	4.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.9	17.9	23.7	20.3	17.2	23.0	20.1	16.9	22.7	70.3	26.0	68.5	25.4	67.0	25.4	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.1	6.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	25.1	26.2	25.2	23.3	22.2	21.3	20.4	20.9	22.2	22.6	22.5	23.4
	DBStd	2.14	1.22	1.42	1.58	1.29	1.04	1.20	1.35	1.26	1.30	1.45	1.06	1.25
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDD10.0	4717	468	453	471	398	378	339	323	337	367	389	376	417
	CDD18.3	1677	210	220	212	148	119	89	66	79	117	131	126	158
	CDH23.3	9623	1894	2193	1710	594	293	178	124	217	479	474	462	1005
CDH26.7	2710	716	912	588	81	11	8	4	8	54	64	43	219	
Wind	WSAvg	3.0	3.2	3.2	2.8	2.3	2.5	3.0	3.8	4.0	3.5	2.6	2.3	2.5
Precipitation	PrecAvg	701	19	20	60	196	134	18	17	14	20	103	97	40
	PrecMax	1259	160	100	182	650	470	56	60	60	85	336	277	161
	PrecMin	356	0	0	0	51	20	1	0	0	0	10	19	0
	PrecStd	204	33	24	52	136	81	14	13	13	23	79	66	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	34.0	33.2	29.6	27.9	27.4	27.1	27.8	29.2	29.8	29.1	31.1
		MCWB	18.0	17.7	17.3	20.3	20.8	18.9	17.9	18.0	18.5	19.7	18.9	17.7
	2%	DB	32.1	32.9	32.0	28.7	26.9	26.1	25.9	26.8	28.0	28.0	28.0	30.0
		MCWB	17.7	17.6	18.4	20.3	20.5	18.5	17.7	17.6	18.3	19.3	19.3	17.7
	5%	DB	31.1	32.1	31.0	27.7	26.0	25.1	24.8	25.7	27.0	27.0	27.1	29.0
		MCWB	17.5	17.5	18.4	20.3	20.3	18.2	17.5	17.3	18.0	19.0	19.5	17.9
	10%	DB	30.3	31.4	30.0	26.8	25.0	24.1	23.7	24.6	26.0	25.9	26.1	27.9
		MCWB	17.4	17.5	18.5	20.5	20.2	18.1	17.4	17.2	17.7	19.1	19.6	18.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.6	21.4	22.2	22.5	22.4	20.3	21.2	20.8	20.2	21.6	22.1	22.0
		MCDB	28.1	28.4	26.7	26.2	25.9	24.6	23.1	22.7	26.3	25.5	25.7	26.6
	2%	WB	20.6	20.5	21.5	21.8	21.6	19.5	19.3	18.7	19.4	20.9	21.4	21.2
		MCDB	26.9	27.6	26.1	25.7	24.8	23.7	22.4	24.0	25.4	24.6	24.8	25.7
	5%	WB	19.9	19.9	20.9	21.4	21.1	19.0	18.4	18.1	18.9	20.5	20.8	20.6
		MCDB	26.1	27.2	25.8	25.2	24.2	23.2	22.3	23.4	24.7	24.1	24.1	24.9
	10%	WB	19.2	19.4	20.5	21.0	20.7	18.6	17.9	17.6	18.3	20.0	20.4	20.1
		MCDB	26.0	26.8	25.5	24.6	23.6	22.6	22.0	22.9	24.2	23.6	23.8	24.2
Mean Daily Temperature Range	5% DB	MDBR	11.3	11.1	9.4	6.9	6.2	6.7	6.3	7.2	8.1	7.2	7.4	9.6
		MCDBR	12.5	11.9	10.9	8.3	7.3	8.0	8.2	8.9	9.5	9.0	9.0	11.0
		MCWBR	4.5	4.0	3.8	3.1	2.9	2.9	3.2	3.1	3.2	3.1	3.5	3.9
	5% WB	MCDBR	10.2	10.5	8.6	7.0	6.3	6.8	6.6	7.4	8.4	7.2	6.9	8.7
		MCWBR	4.1	4.2	3.3	3.2	3.0	2.8	3.6	3.2	3.4	3.2	3.4	4.2
Clear-Sky Solar Irradiance	taub	0.338	0.357	0.379	0.380	0.381	0.447	0.454	0.406	0.381	0.373	0.367	0.347	
	taud	2.558	2.490	2.455	2.504	2.504	2.238	2.204	2.369	2.449	2.515	2.519	2.552	
	Ebn at Noon	986	970	942	924	902	830	829	888	927	942	949	971	
	Edh at Noon	104	114	118	110	107	137	143	124	117	110	108	103	
All-Sky Solar Radiation	RadAvg	6.54	6.76	6.41	5.38	5.07	4.51	4.43	5.01	5.65	5.03	5.29	5.98	
	RadStd	0.34	0.31	0.41	0.25	0.20	0.37	0.33	0.27	0.24	0.21	0.22	0.42	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page